The end product of the neptunium series is ^{209}Bi, which is the only stable isotope of bismuth. Seven α- and four β-decays are required in the sequence from the parent ^{237}Np to ^{209}Bi. An important nuclide in the neptunium decay series is the uranium isotope ^{233}U, which has a half-life of 1.59×10^5 y (the most stable intermediate) and, like ^{235}U, is fissionable by slow neutrons.

The long-lived plutonium isotope ^{244}Pu (belonging to the $4n$ series; see also Fig. 14.1), which decays through α-emission and spontaneous fission (0.13%) with a total half-life of 8.26×10^7 y, was discovered in rare earth minerals in 1971. If this is a survival of primeval ^{244}Pu, only 10^{-15} % of the original can remain. An alternate possibility is that this ^{244}Pu is a contaminant from cosmic dust (e.g. from a supernova explosion in more recent times than the age of the solar system).

13.6. RADIUM AND RADON IN THE ENVIRONMENT

In uranium ore radioactive equilibrium is established between the mother ^{238}U and daughters in the decay chain (see Fig. 1.2, $4n + 2$ series) at *the rate of the daughter half-life* and at the *level of the mother decay rate*. The chain passes ^{226}Ra and ^{222}Rn and daughters down to ^{206}Pb at a rate corresponding to the original amount of ^{238}U, as long as the host material is undisturbed. The equilibrium between ^{226}Ra and ^{210}Pb is established within some weeks. Since ^{222}Rn is a gaseous intermediate, its daughters are likely to be formed in air ("radon daughters"). Radon diffuses out of thorium and uranium minerals, and adds radioactivity to the ground water and to the atmosphere both by its own presence and that of its daughters. Since Ra and Rn are among the most radio-toxic substances existing, causing bone and lung cancer at relatively low concentrations (the risk levels are discussed in Ch. 15), special attention must be devoted to their appearance in nature.

Common Rn-concentrations in ground water are 5–300 kBq/m^3, but in areas of U-rich granite values $\gg$1 MBq ^{222}Rn/m^3 occur. Tap water usually contains of the order of 1 kBq/m^3. In many places water from hot mineral wells is considered beneficial to health both for bathing and for drinking ("spas" or hot springs). The water may be warm due to radiogenic heating at the source (minerals rich in U or Th) and have a high content of dissolved radium and radon. Thus, in famous spas in Europe the ^{222}Rn concentration for "therapeutic inhalation" may be 1 MBq/m^3 air (Baden-Baden). From the Joachimsthal U-mine in Bohemia, which contains a number of hot wells (29°C), water containing 10–15 MBq ^{222}Rn/m^3 is pumped to spas, where it is used as medical treatment against rheumatism (a 30 min bath a day).

The average exhalation rate of radon from the ground is 5–50 mBq/m^{2}s, leading to a near ground level radon concentration of 1–10 Bq/m^3, but varies widely with ground conditions. In soil over Swedish uranium bearing shale ($\approx$300 ppm U) the ^{222}Rn concentration can exceed 1 MBq/m^3, though on the average the concentration in Scandinavian air (the main geology consists of granitic rocks) is only 3 Bq/m^3.

Representative values are for the US 0.1–10 Bq/m^3, UK and Germany ~3, and 10 (average) for France. The concentration in air above ground depends also on temperature and wind conditions.

Many ores contain small amounts of uranium. During processing, uranium and/or its daughters may enter the product, causing a radioactive contamination problem. For example, when apatite is used to produce phosphoric acid, the gypsum by-product contains all the radium originally present, producing a γ-ray and inhalation hazard from Rn-daughters, making it unsuitable for building material.

Radon concentrations in indoor air may be quite high, depending on site and building material. The ^{226}Ra content e.g. in German building materials varies from >500 (gypsum) to 60 (brick) Bq/kg; slag used in Poland contains <800, Italian tuff ≈280, concrete in Hungary ≈13, and white bricks in the UK only ≈4 Bq/kg. The indoor concentration of radon also depends on the way in which the house is built and used (poorly ventilated, etc). In the US it varies between <1 to >1000 Bq/m^3. Authorities no longer recommend very tight houses, as suggested in the 1970's to reduce heating costs. At levels <70 Bq/m^3 the Rn-hazard is considered negligible. In Sweden (pop. 8 M people) 50% of the houses have 70–200 Bq/m^3, while 40 000 houses have been classified as "Rn-houses", i.e. their indoor concentration of Rn plus daughters exceed 400 Bq/m^3. If the ground is the main source of radon, ventilation of the basement may be sufficient to declassify a "Rn-house".

Even coal contains small amounts of uranium, 4 to 300 kBq/ton, a typical value being 20 kBq/ton. When coal is burnt, the more volatile U-daughter products are released into the atmosphere, while the less volatile ones are deposited in the dust filters. A 1 GW$_e$ coal-fired power plant typically releases 60 GBq ^{222}Rn and 5 GBq ^{210}Pb+Po annually, while ~3 MBq daughter products are obtained per ton of fly ash.

13.7. DISEQUILIBRIUM

Uranium ores at surface level are usually identified from the penetrating γ-ray emissions of the daughter products. However, it has been observed that some peat, which absorb uranium from local drainage, contain very little of the daughter products because they have been formed so recently that radioactive equilibrium has not been established.

The decay products of uranium passes over 10 elements (Fig. 1.2), all with very different chemical properties. These elements are transported by groundwater (*migrate*), the solute composition of which varies with the surrounding rock/soil minerals. The different elements migrate at different rates due to their different chemistries, dissolving in some areas and precipitating in others. If the mother and daughter in a radioactive chain migrate at different velocities during a time which is short in relation to the daughter half-life, the radioactive equilibrium is disturbed; this is referred to as *disequilibrium*. Such disequilibrium can be used to measure the age of the sample.